



Survey Methods

4.1 - Non-response bias

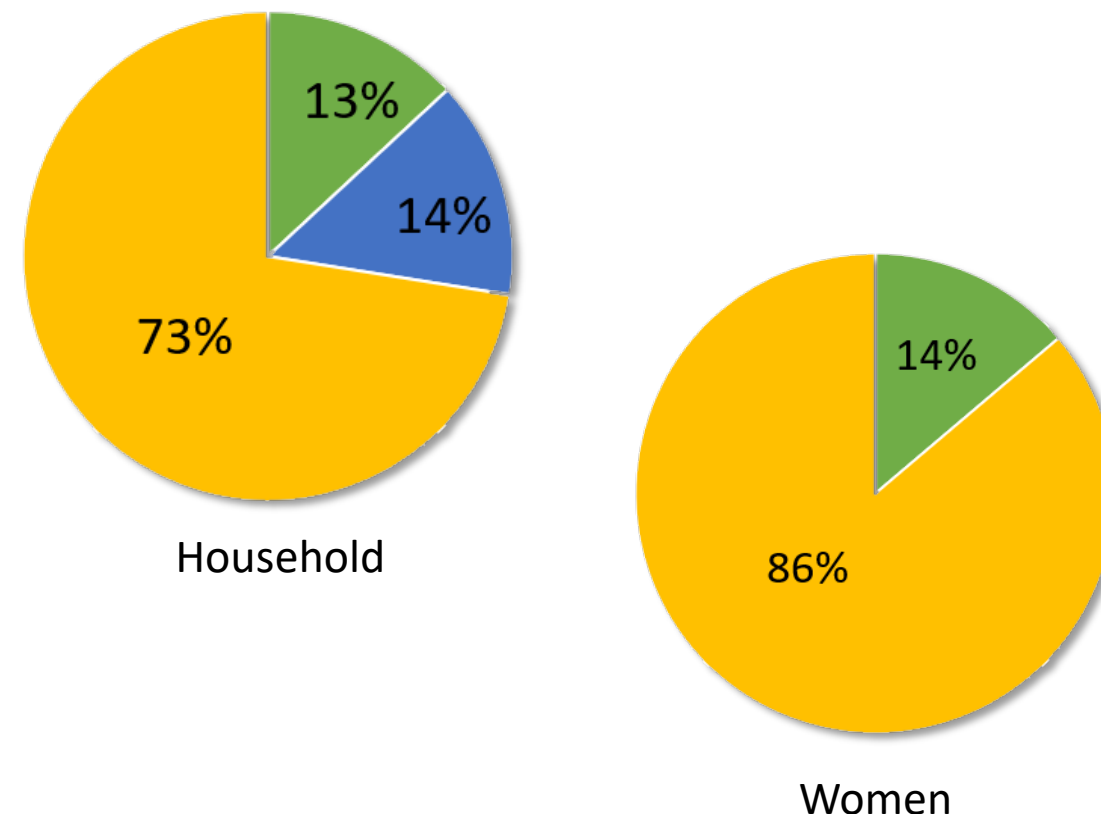
Dr Annibale Cois, MEng, MPH, PhD
Division of Health Systems and Public Health
acois@sun.ac.za

Outline

- Design vs realised sample
- Limiting non-response bias
- Adjusting for non-response bias

Design vs realised sample

Result	Total
Household interviews	
Households selected	15,292
Households occupied	13,288
Households interviewed	11,083
Household response rate ¹	83.4
Interviews with women age 15-49	
Number of eligible women	9,878
Number of eligible women interviewed	8,514
Eligible women response rate ²	86.2
Household interviews in subsample selected for male survey and adult health module	
Households selected	7,623
Households occupied	6,590
Households interviewed	5,477
Household response rate ¹	83.1
Interviews with men age 15-59	
Number of eligible men	4,952
Number of eligible men interviewed	3,618
Eligible men response rate ²	73.1
Interviews with adults age 15+	
Number of eligible adults	12,717
Number of eligible adults interviewed	10,336
Eligible adults response rate ²	81.3



Total: $0.73 * 0.86 = 0.63 = 63\%$

¹ Households interviewed/households occupied

² Respondents interviewed/eligible respondents

Table 1.1 Results of the household and individual interviews

SADHS 2016

(National Department of Health and ICF 2019:9)

Design vs realised sample

Result	Total	Residence	
		Urban	Non-urban
Household interviews			
Households selected	15,292	9,547	5,745
Households occupied	13,288	8,397	4,891
Households interviewed	11,083	6,556	4,527
Household response rate ¹	83.4	78.1	92.6
Interviews with women age 15-49			
Number of eligible women	9,878	5,858	4,020
Number of eligible women interviewed	8,514	4,805	3,709
Eligible women response rate ²	86.2	82.0	92.3
Household interviews in subsample selected for male survey and adult health module			
Households selected	7,623	4,751	2,872
Households occupied	6,590	4,164	2,426
Households interviewed	5,477	3,240	2,237
Household response rate ¹	83.1	77.8	92.2
Interviews with men age 15-59			
Number of eligible men	4,952	2,996	1,956
Number of eligible men interviewed	3,618	2,021	1,597
Eligible men response rate ²	73.1	67.5	81.6
Interviews with adults age 15+			
Number of eligible adults	12,717	7,463	5,254
Number of eligible adults interviewed	10,336	5,685	4,651
Eligible adults response rate ²	81.3	76.2	88.5

¹ Households interviewed/households occupied

² Respondents interviewed/eligible respondents

Low response rates →
reduced sample size →
reduced power/precision

Unequal response rates →
Bias

Table 1.1 Results of the household and individual interviews

SADHS 2016

(National Department of Health and ICF 2019:9)

Example (hypothetical)

Background characteristic	Reads a newspaper at least once a week
Urban	49.8
Non-urban	20.9

%

60% of the population is urban

‘Truth’ for the South African population

True proportion of South African women who read a newspaper at least once a week



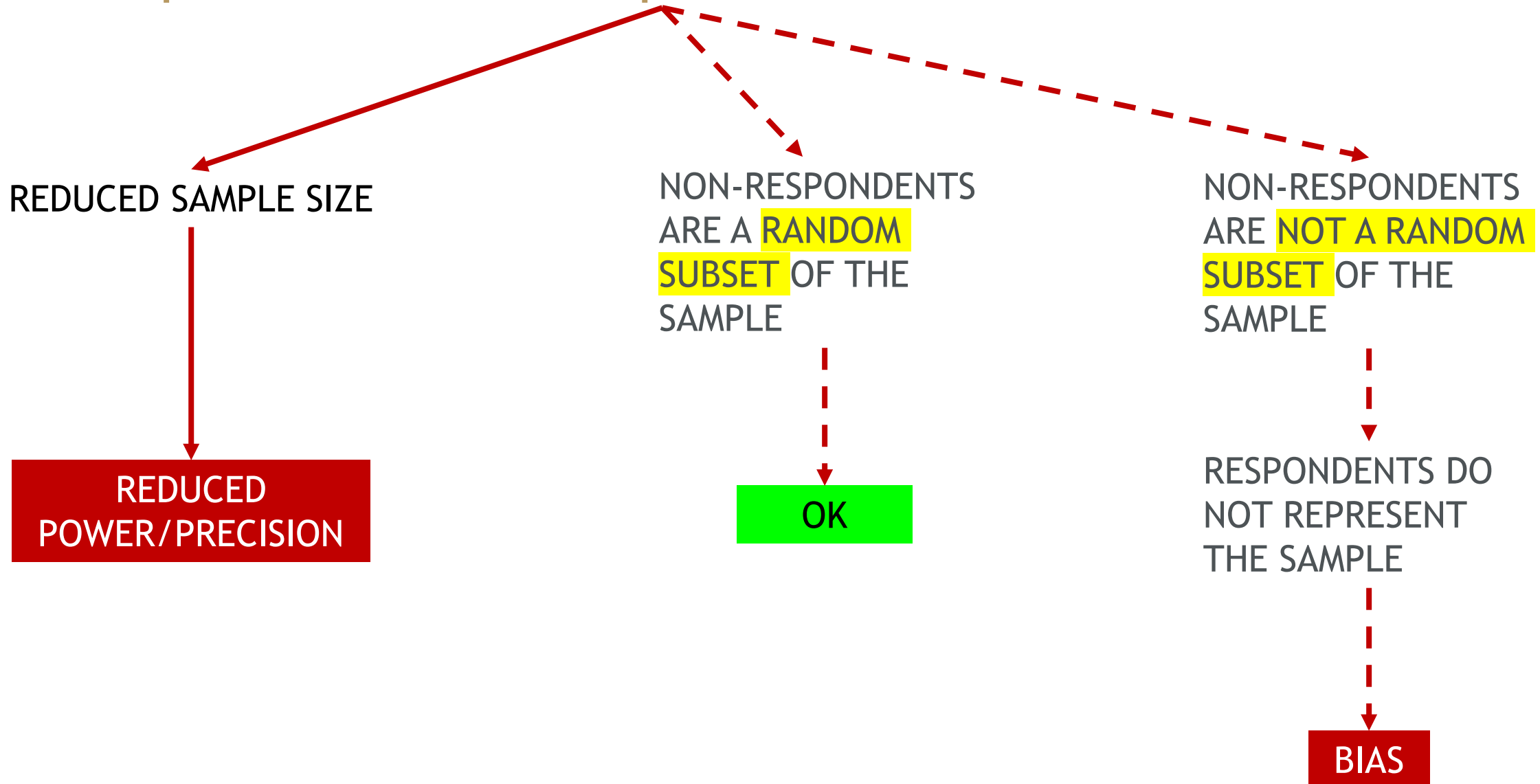
$$(49.8\% \times 0.60 + 20.9\% \times 0.40) / (0.60 + 0.40) = 38.2\%$$

Proportion of South African women who read a newspaper at least once a week according to the survey (before adjustment)



$$49.8\% \times 0.60 \times 0.781 \times 0.820 + 20.9\% \times 0.40 \times 0.926 \times 0.923 / (0.60 \times 0.781 \times 0.820 + 0.40 \times 0.926 \times 0.923) = 36.2\%$$

Consequences of non-response



Example of precision reduction

Sample size for estimation of proportions:

$$N = \frac{Z_{\alpha/2}^2 \times P \times (1 - p) \times D}{E^2}$$



$$E = \sqrt{\frac{Z_{\alpha/2}^2 \times P \times (1 - p) \times D}{N}}$$

Annotations for the margin of error formula:

- 50% (pointing to P)
- 1.96 ($\alpha = 0.05$) (pointing to $Z_{\alpha/2}$)
- 1 (simple random sampling) (pointing to D)
- $N = 1000$ (pointing to N)

Response rate = 1

Response rate = 0.8

Response rate = 0.6

Response rate = 0.5

Precision = ± 3.1

Precision = ± 3.5

Precision = ± 4.0

Precision = ± 4.4

HIV-prevalence estimates form SABBSSM V: precision

	Eligible	Tested (%)	HIV-positive (%)	Std. Err.		CV (%)	Design effects (DEFT)	Square root of the design effect (DEFF)
				SRS	Complex			
Sex								
Male	17 683	57.7	10.6	0.003	0.005	4.9	1.8	3.3
Female	21 273	64.3	17.3	0.003	0.006	3.6	1.8	3.3
Age (years)								
0–14	11 828	56	2.7	0.002	0.003	11.0	1.5	2.4
15–24	6 854	66.9	7.9	0.004	0.006	7.5	1.4	2.0
25–49	12 607	61.5	26.4	0.005	0.009	3.5	2.0	4.0
50 and older	7 685	64.4	12.5	0.005	0.009	7.1	1.7	2.7
25 and older	20 292	62.6	22.2	0.004	0.008	3.6	2.2	4.7
15–49	19 461	63.4	20.6	0.004	0.007	3.5	2.0	3.9
Race								
Black African	25 837	65.4	16.6	0.003	0.005	3.1	1.9	3.8
White	2 555	42.3	1.1	0.002	0.007	61.9	2.8	7.9
Coloured	7 197	61.7	5.3	0.005	0.007	13.3	1.4	2.1
Indian/Asian	3 369	42.6	0.8	0.004	0.003	41.6	0.9	0.9

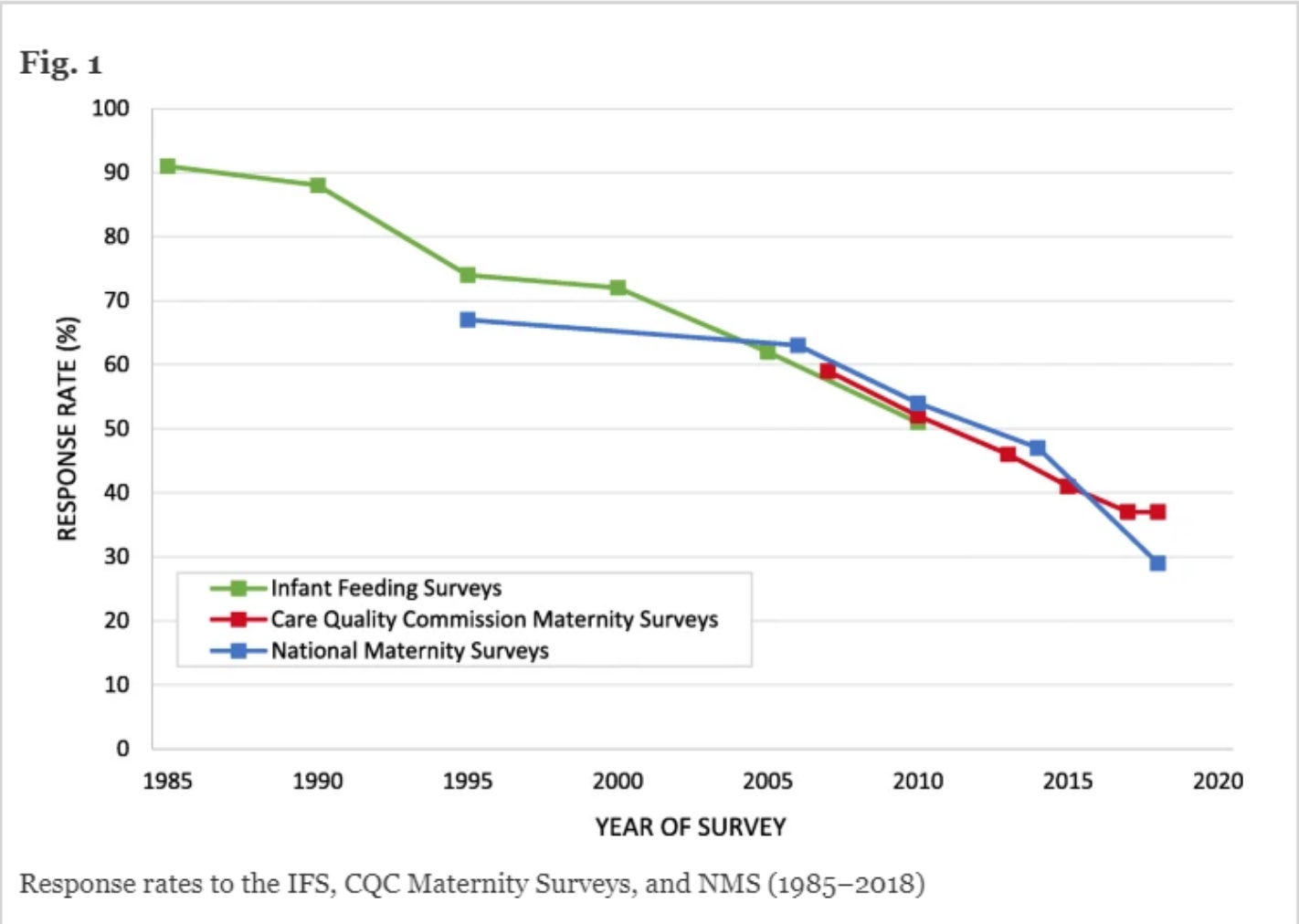
(Table adapted from Simbayi et al. 2019:165)

Design vs realised sample: many reasons for non-response

The survey targeted 15 000 VPs but only 12 435 of these (82.9%) were approached. Among the 12 435 VPs, only 11 776 (94.7%) were valid VPs (see Table 3.2). Of all the VPs approached, 5.6% (659) were invalid or clearly abandoned households. An additional 813 VPs were not realised for other reasons, as mentioned earlier. Of the 11 776 valid VPs, 9 656 agreed to participate in the survey at the household level. This translates to a household response rate of 82%. The proportions of non-responses at household level were as follows:

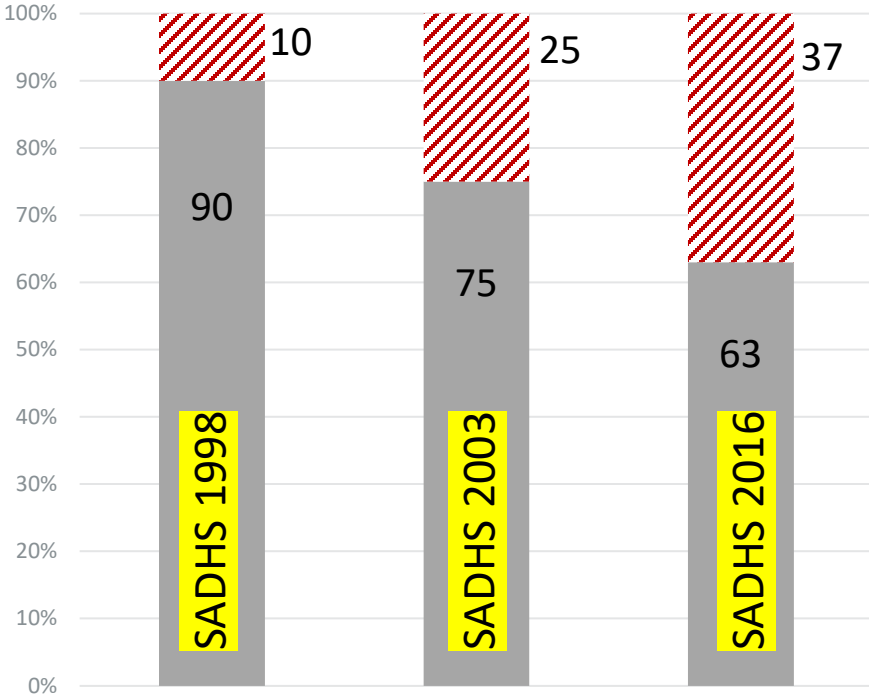
- A total of 1 307 of 11 776 (11.1%) households refused to take part in the survey.
- A total of 813 of 11 776 (6.9%) were valid VPs but the households were found to be empty after the required four repeat visits to interview the head of the household, or could not be considered for other reasons.

Survey response rates are declining...



(Figure from: Harrison et al. 2020)

Response rates (HH*Individual) for women in reproductive age in the SADHS surveys



Where Have the Respondents Gone? Perhaps We Ate Them All FREE

Thomas J Leeper ✉

Public Opinion Quarterly, Volume 83, Issue S1, 2019, Pages 280–288,

<https://doi.org/10.1093/poq/nfz010>

Published: 19 June 2019

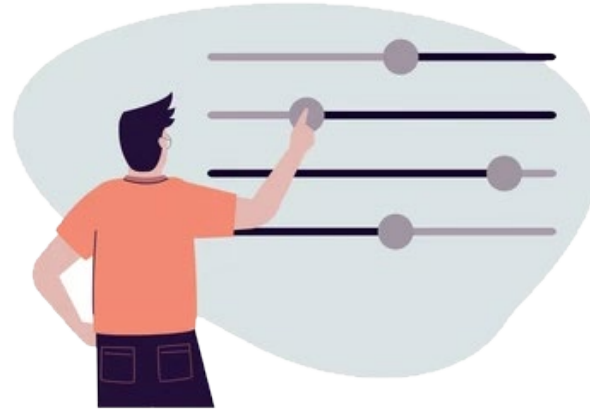


What to do?

PREVENT



ADJUST



THINK



Prevention



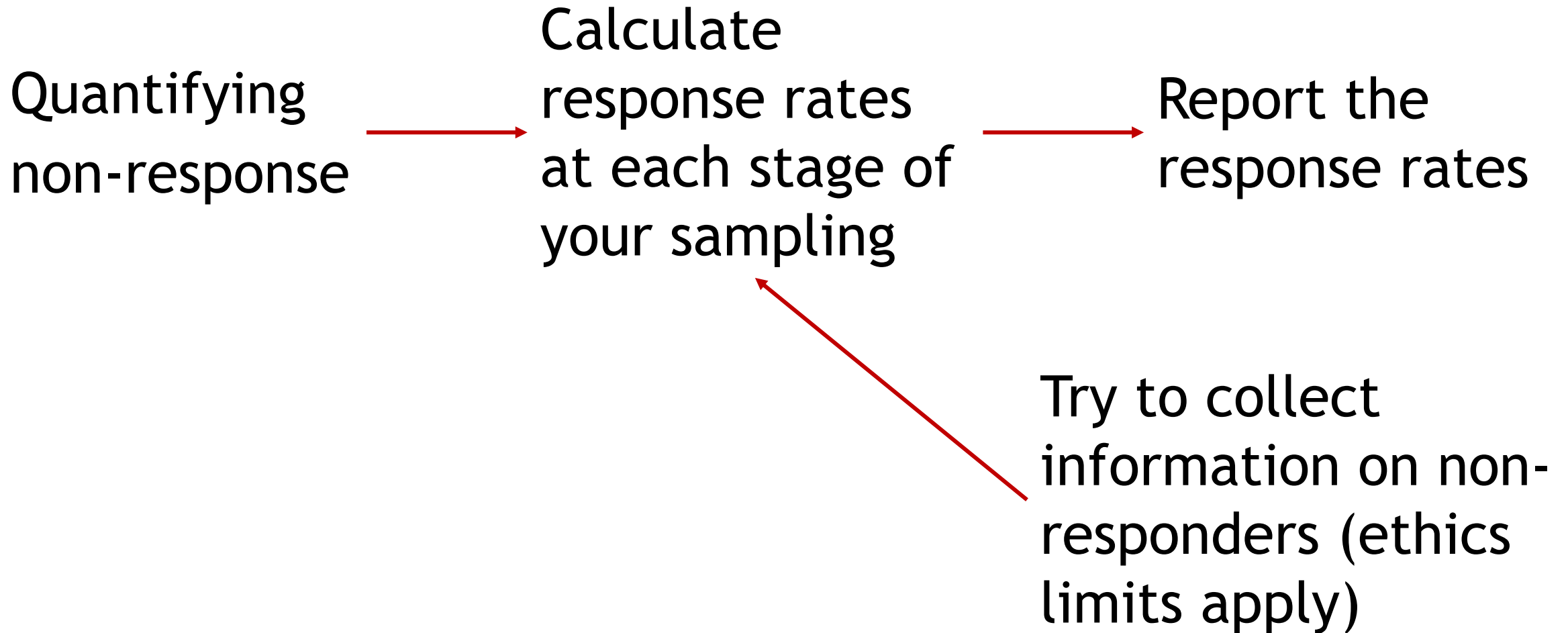
- Understanding and Reducing Respondent Burden
- Incentives
- New data collection modes, multiple modes
- New frames and methods of sampling
- Responsive Design



Don't let a bad survey ruin your customer's last impression.



- Quantifying non-response
- Sensitivity analysis
- Speculate on possible consequences



Example: HIV prevalence estimates in SABSSM V

HIV Indicator		Females	95% CI ²	Males	95% CI	Total	95% CI
HIV Prevalence (%)							
Age	0-14 years	3.0	2.3-3.9	2.4	1.8-3.2	2.7	2.2-3.3
	15-49 years	26.3	24.5-28.2	14.8	13.3-16.5	20.6	19.2-22.0
	50-64 years	18.7	15.7-22.0	15.3	12.2-19.2	17.2	14.9-19.8
	15-64 years	25.0	23.2-26.8	14.9	13.5-16.4	20.0	18.7-21.4
Race ⁴	Black African	20.6	19.3-22.0	12.5	11.4-13.7	16.6	15.6-17.7
	White	0.9	0.2-3.3	1.3	0.3-4.7	1.1	0.3-3.6
	Coloured	5.8	4.2-8.1	4.7	3.2-6.8	5.3	4.0-6.8
	Indian/Asian	1.3	0.4-3.7	0.4	0.2-1.1	0.8	0.4-1.9

(Table adapted from Simbayi et al. 2019:167)

Table 3.4: HIV-risk associated characteristics among respondents aged 15 years and older who were interviewed and either provided or did not provide a blood sample, South Africa, 2017

Demographic or risk characteristic	Interviewed and tested		Interviewed but refused to provide blood sample		p-value
	n	%	n	%	
Perceived risk of getting HIV					
At risk	12 413	85.5	6 328	89.1	<0.001***
Not at risk	2 113	14.6	776	10.9	0.012**
Total	14 526	100	7 104	100	
Ever had an HIV test					
No	4 048	25.6	2 204	29.8	<0.001***
Yes	11 759	74.4	5 197	70.2	<0.001***
Total	15 807	100	7 401	100	
Number of partners in last 12 months					
One sexual partner	7 550	90.5	3 426	92.7	<0.001***
Two sexual partners	501	6.0	173	4.7	0.518
Three or more sexual partners	295	3.5	97	2.6	0.664
Total	8 346	100	3 696	100	(Tab

(Table adapted from Simbayi et al. 2019:43-45)

- Do you think that these differential responses rates could have affected the estimates of HIV prevalence in SA?
- How and why?



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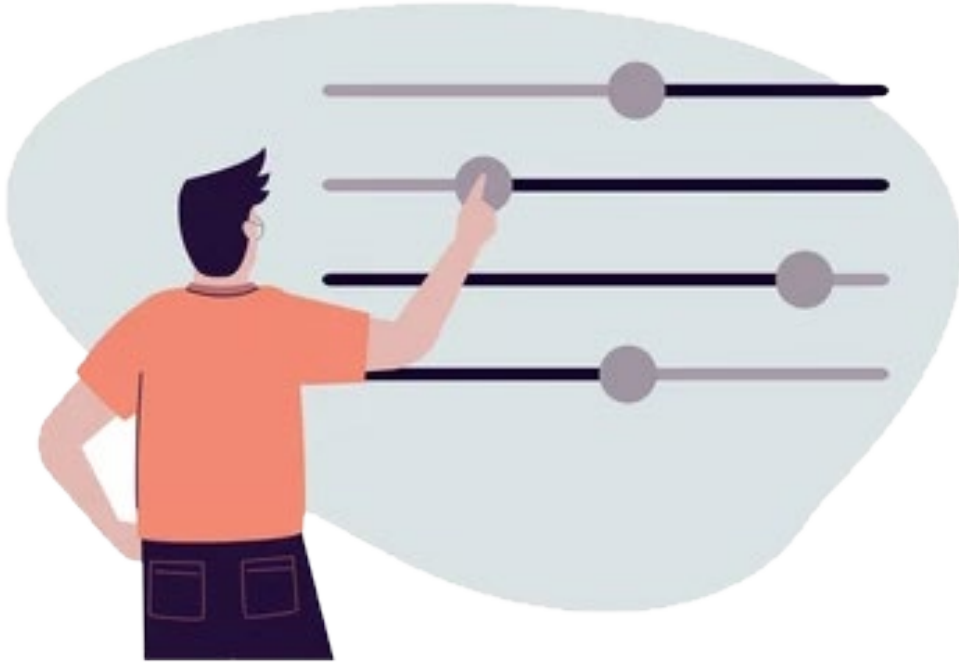
Bias?



(Table adapted from Simbayi et al. 2019:43-45)



this is what goes in the “strengths and limitations” section of your study report!



- Weighting
- Additional data for adjustment
- Modelling

Weighting

You know this, for example, from census



Weighting

Calculate weights



in the calculations...

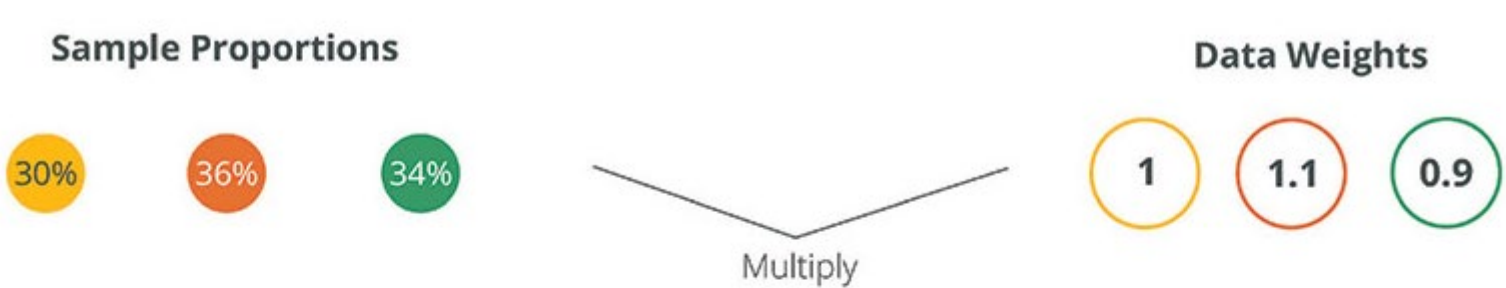
1 element in the sample is counted as 1 element of the population.

1 element in the sample is counted as 1.1 elements of the population.

1 element in the sample is counted as 0.9 elements of the population.

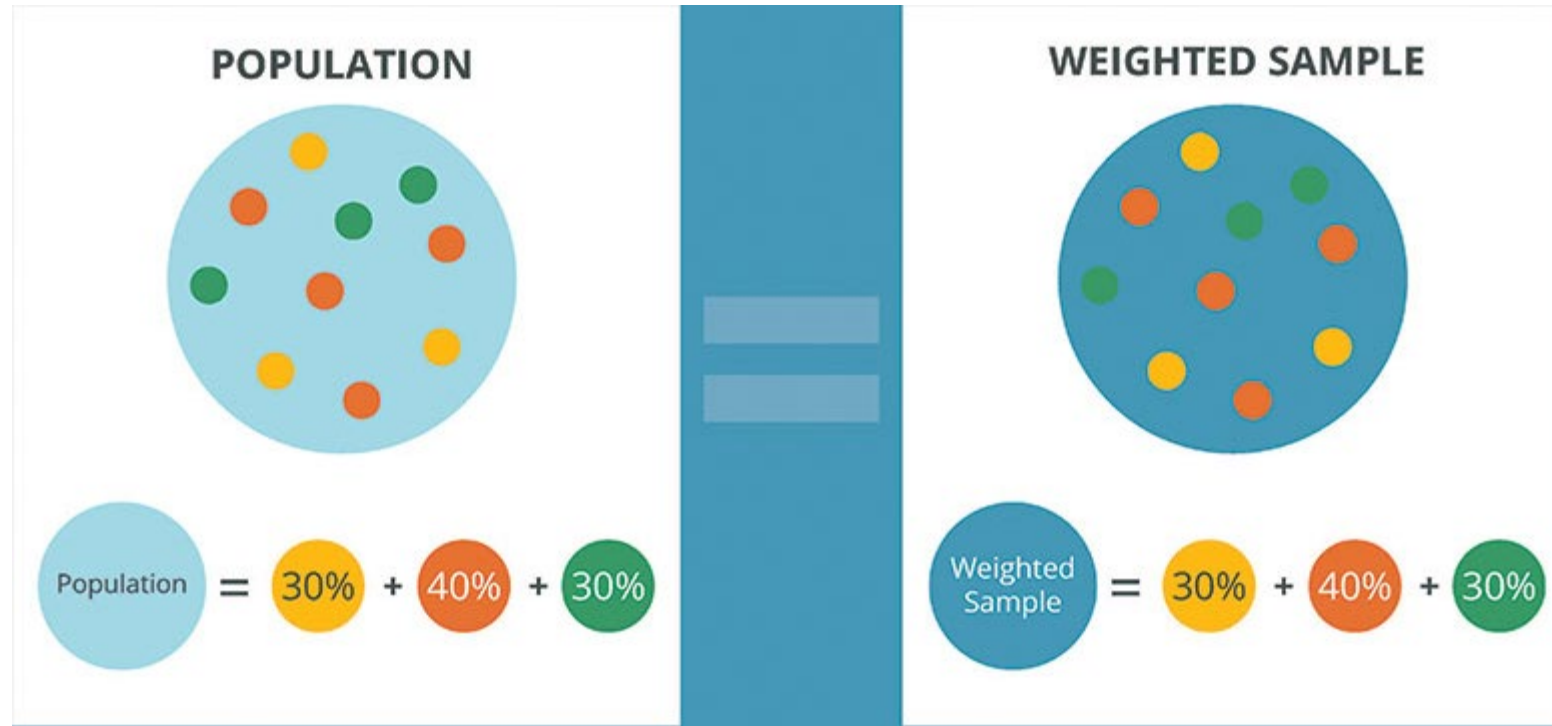
Weighting

Use weights



Age group	15-24	25-34	35+
Calculation	$30\% \times 1 = 30\%$	$36\% \times 1.1 = 40\%$	$34\% \times 0.9 = 30\%$
Weighted Sample Proportions	30%	40%	30%

Weighting



Proportion of people living in urban areas in the sample:
60% of the population is urban

Background characteristic	Reads a newspaper at least once a week
Urban	49.8
Non-urban	20.9

%

True proportion of South African women who read a newspaper at least once a week



38.2%

N = 1000 → 600 Urban, 400 rural

Design sample: 100

100% response rate

Response rate < 100%

n = 100 → 60 urban, 40 rural

Proportion readers = $(60 * 49.8\% + 40 * 20.9\%) / 100 =$

38.2% → correct!

n = $60 * 0.781 * 0.820$ urban, $40 * 0.926 * 0.923$ rural

n = 38 urban, 34 rural = 72

Proportion readers = $(38 * 49.8\% + 34 * 20.9\%) / 72 =$

36.2% → wrong!

Weighting

Urban in population: 60%

Urban in sample: $38/72 = 52.8\%$

$$\text{Weight 1:} \\ 60/52.8 = 1.136$$

Rural in population: 40%

Rural in sample: $34/72 = 47.2\%$

$$\text{Weight 2:} \\ 40/47.2 = 0.847$$

$$\text{Proportion readers} = (38 * 49.8\% * 1.136 + 34 * 20.9\% * 0.847) / 72 \\ =$$

38.2% → correct!

Some notes:

Weighting is essential when response rates are sub-optimal and differential across population strata and the characteristic of interest varies across strata

Weighting is also used when the over/undersampling is by design

Weighting reduces bias (but might also reduce precision)

Weighting requires external knowledge of the population totals/proportions (marginals)

Weighting is normally done in order to match more than one characteristics of the population (often: age group * sex * area)

Weighting in cases of multiple variables involved and/or for complex sampling schemes is complex (and requires statistical support).

Example

Technical papers



<http://www.nids.uct.ac.za/publications/technical-papers>

Summary questions

- What are the potential consequence of non-response?
- What can a research do to limit the negative consequences of non-response?
- Reporting on non-response

References

Fowler Jr, Floyd J. 2013. *Survey Research Methods*. Sage publications. Chapter 4

Leeper, Thomas J. 2019. 'Where Have the Respondents Gone? Perhaps We Ate Them All'. *Public Opinion Quarterly* 83 (S1): 280–88. <https://doi.org/10.1093/poq/nfz010>.

Simbayi, LCZK, Khangelani Zuma, Nompumelelo Zungu, S Moyo, E Marinda, S Jooste, M Mabaso, S Ramlagan, A North, and J Van Zyl. 2019. 'South African National HIV Prevalence, Incidence, Behaviour and Communication Survey, 2017: Towards Achieving the UNAIDS 90-90-90 Targets'. Chapter 2 & pages 39-46.

Other resources

Harrison, Siân, Fiona Alderdice, Jane Henderson, Maggie Redshaw, and Maria A. Quigley. 2020. 'Trends in Response Rates and Respondent Characteristics in Five National Maternity Surveys in England during 1995–2018'. *Archives of Public Health* 78 (1): 46. <https://doi.org/10.1186/s13690-020-00427-w>.

National Research Council. 2013. *Nonresponse in Social Science Surveys: A Research Agenda*. <https://doi.org/10.17226/18293>.



Thank you

Annibale Cois

Education Building, 4th floor, Room 4060B

acois@sun.ac.za